

NETWORK TOPOLOGIES IN COMPUTER NETWORKS

Presented By

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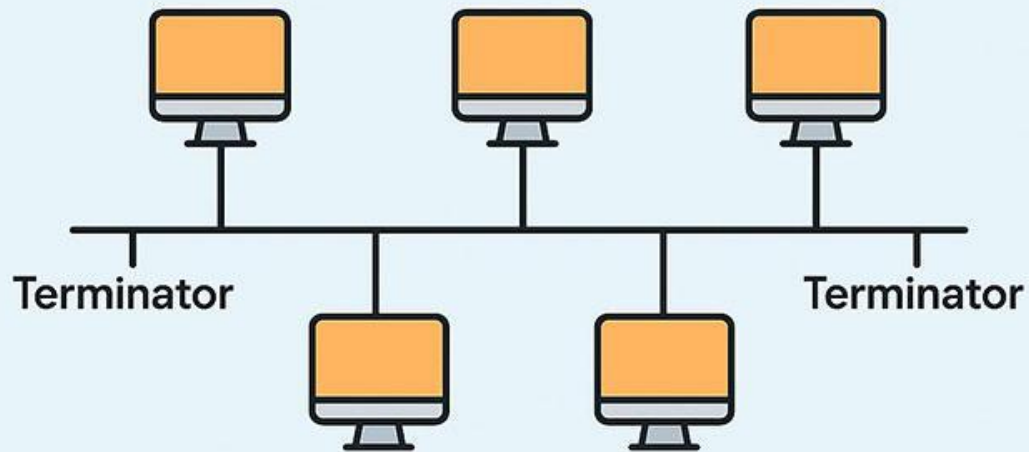
Course Name: Computer Networks

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Introduction to Network Topology

- ▶ **What is Network Topology?**
- ▶ A **network topology** is the physical or logical arrangement of computers, networking devices, and communication links in a network. It determines how devices are connected and how data travels between them.
- ▶ **Why is it Important?**
- ▶ Provides an organized network structure.
- ▶ Improves communication efficiency.
- ▶ Makes troubleshooting easier.
- ▶ Helps in selecting the best network design based on requirements.

Bus Topology



Bus Topology

Definition

- All devices are connected to a single communication cable called the **backbone**.

Advantages

- Low cost
- Easy to install
- Requires less cable

Disadvantages

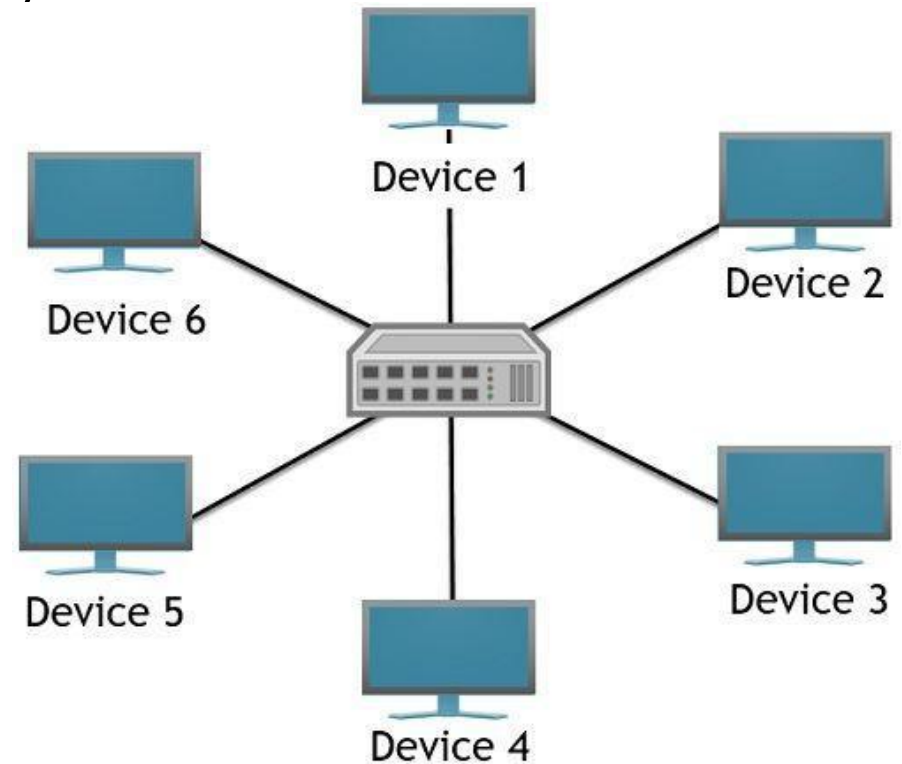
- Backbone failure stops the entire network
- Difficult to troubleshoot
- Performance decreases as more devices are added

Applications

- Small office networks
- Temporary LAN setups

Star Topology

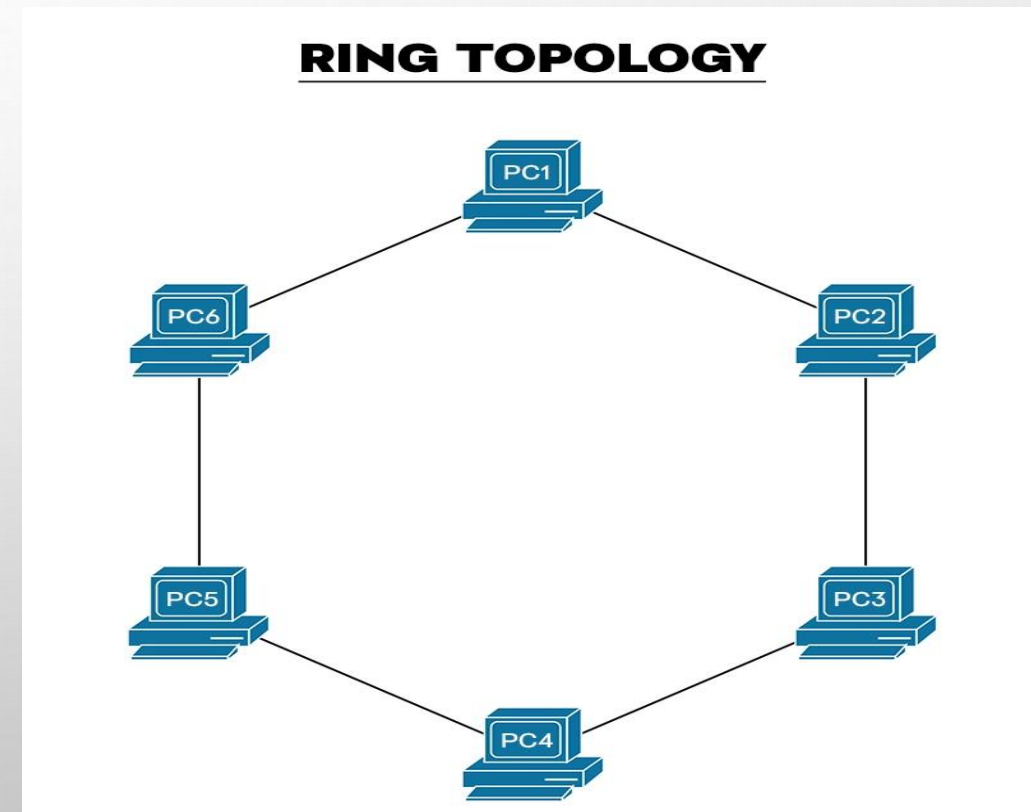
- **Definition**
- Every device connects to a central device (switch or hub).
- **Advantages**
- Easy to manage
- Failure of one cable affects only one device
- High performance
- **Disadvantages**
- Central switch failure affects the whole network
- More cabling required
- **Applications**
- Schools
- Offices
- Home Wi-Fi networks



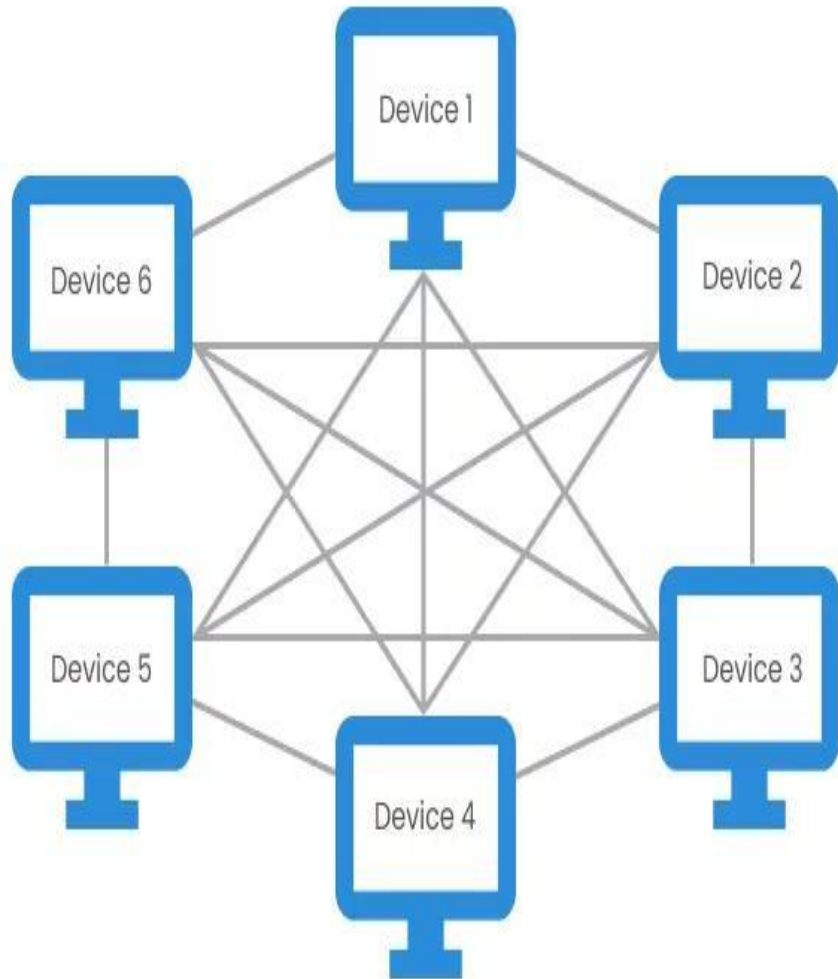
Star Topology

RING TOPOLOGY

- **DEFINITION**
- EACH DEVICE IS CONNECTED TO EXACTLY TWO NEIGHBORING DEVICES, FORMING A CLOSED LOOP.
- **ADVANTAGES**
- EQUAL ACCESS FOR ALL DEVICES
- NO DATA COLLISIONS
- **DISADVANTAGES**
- FAILURE OF ONE NODE MAY AFFECT THE ENTIRE NETWORK
- DIFFICULT TO EXPAND
- **APPLICATIONS**
- FIBER OPTIC COMMUNICATION
- SOME METROPOLITAN NETWORKS



Mesh Topology



- ◇ **Definition**

- ◇ Each device is connected to many or all other devices.

- ◇ **Advantages**

- ◇ Highly reliable

- ◇ Multiple paths for data

- ◇ Excellent fault tolerance

- ◇ **Disadvantages**

- ◇ Expensive

- ◇ Complex installation

- ◇ Large amount of cabling

- ◇ **Applications**

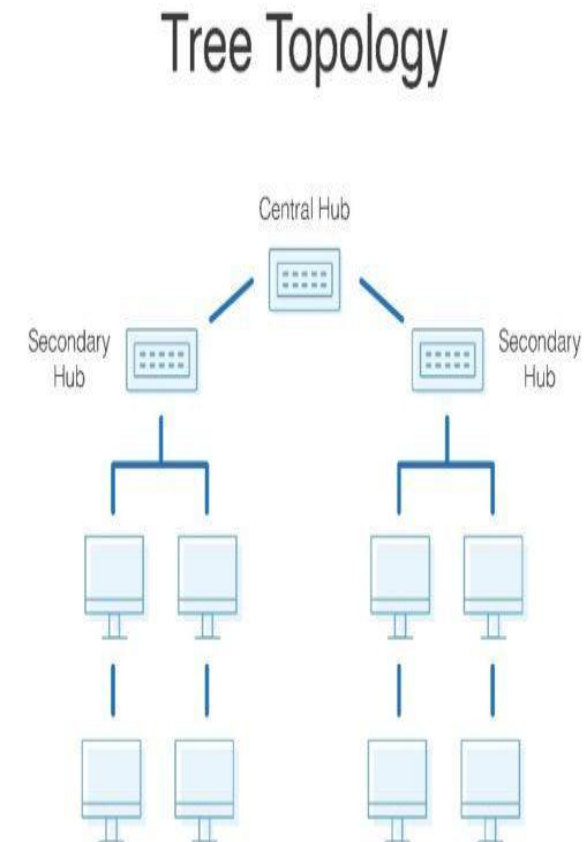
- ◇ Internet backbone

- ◇ Military communication

- ◇ Data centers

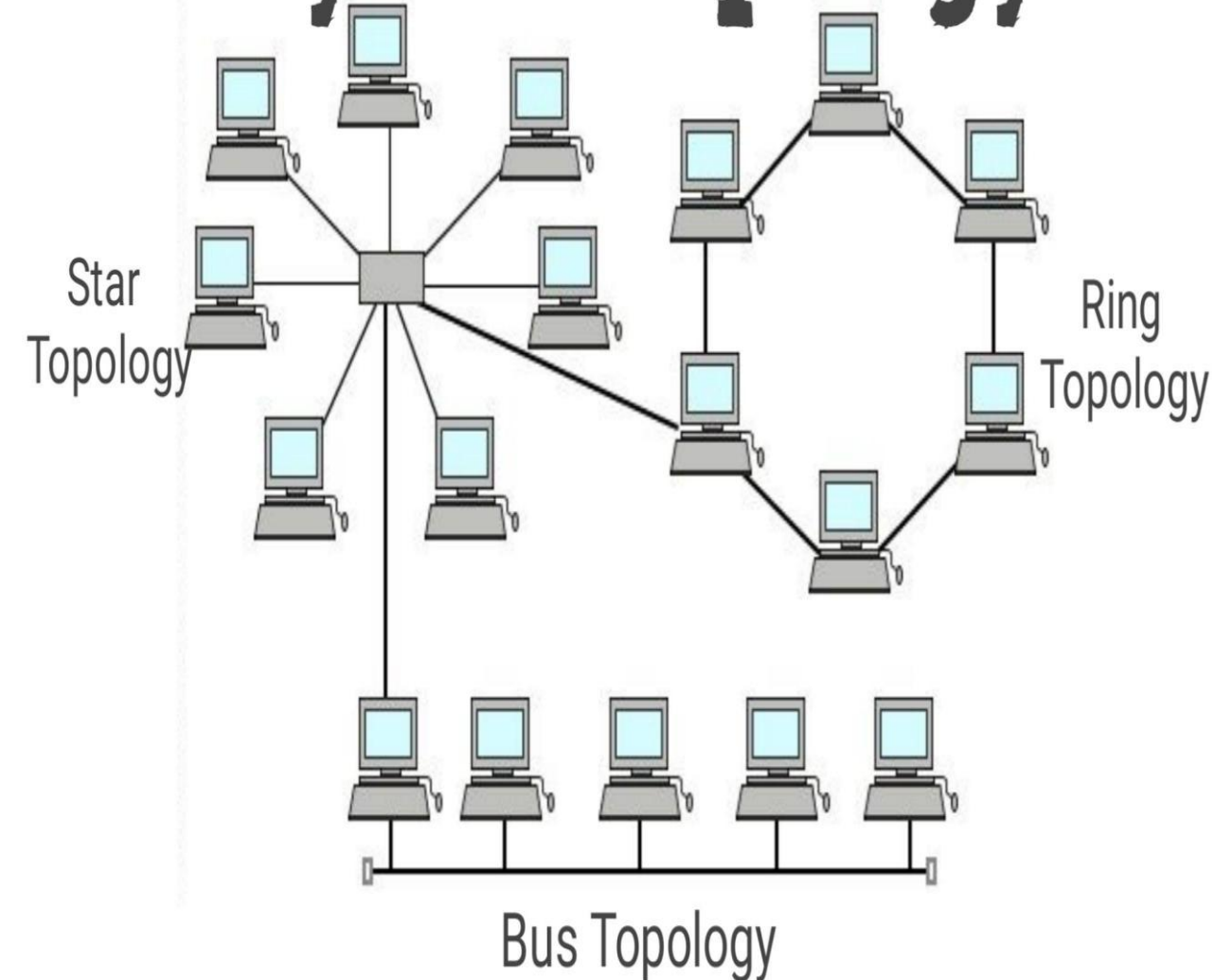
TREE TOPOLOGY

- **Definition**
- A hierarchical combination of multiple star networks connected through a backbone.
- **Advantages**
- Easy to expand
- Supports many devices
- Easy fault isolation
- **Disadvantages**
- Backbone failure affects large portions of the network
- More expensive than star topology
- **Applications**
- Universities
- Large organizations



Hybrid Topology

Hybrid Topology



Definition

A combination of two or more network topologies.

Advantages

Flexible

Scalable

Reliable

Disadvantages

Complex design

High implementation cost

Applications

Banks

Corporate offices

Large enterprises

Types of Network Topologies

- ▶ The six main types of network topologies are:
- ▶ **Bus Topology** – All devices are connected to a single backbone cable.
- ▶ **Star Topology** – All devices are connected to a central hub or switch.
- ▶ **Ring Topology** – Devices are connected in a circular path.
- ▶ **Mesh Topology** – Every device is connected to every other device.
- ▶ **Tree Topology** – A hierarchical combination of Star and Bus topologies.
- ▶ **Hybrid Topology** – A combination of two or more different topologies.

Conclusion

- Network topology defines how devices are connected and communicate.
- Each topology has its own advantages, disadvantages, and applications.
- **Star Topology** is the most commonly used because it is reliable, easy to manage, and simple to expand.
- Choosing the right topology improves network performance, reliability, and scalability.

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P r e s e n t a t i o n T e m p l a t e